

Sushant Gautam | PhD Student — Multimodal Understanding

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Work Experience

Simula Metropolitan Center for Digital Engineering (SimulaMet)

Oslo, Norway

Ph.D. Student (Full-time), Affiliated with *Oslo Metropolitan University*

May 2023 – Present

Conducting research on **multimodal learning for robust and trustworthy vision-language models**; supervising master's students in AI and medical imaging; representing Simula at the NORA Student Council and organizing academic and outreach events; participating in Simula Academy outreach programs to inspire and attract master's students to research.

The Alan Turing Institute

London, United Kingdom

Ph.D. Enrichment Student for the Turing Enrichment Scheme 2024/25

Oct 2024 – Jul 2025

Selected through the Norwegian Artificial Intelligence Research Consortium (NORA); undertook a six-month research stay in London as part of doctoral training; served as Student Representative for the cohort; contributed to discussions that led to the extension of Kvasir-VQA into a multi-task and a multi-complexity benchmark for medical reasoning in gastrointestinal endoscopy; participated in Turing-Roche Co-working sessions, AI-UK 2025 volunteering, and professional development workshops; fostered interdisciplinary collaborations across UK research labs and universities.

Thapathali Campus, Tribhuvan University

Kathmandu, Nepal

Adjunct Lecturer (Contract-based)

Oct 2022 – May 2023

- Lectures and labs for "A Practical Approach to Deep Learning" course.
- Supervised MS projects related to deep learning.

North Star Developer's Village

Lalitpur, Nepal

Engineering Manager

Apr 2019 – May 2023

- ML face verification; pose-based exam cheating prevention; image/audio annotation; ML training platform.
- Semantic essay ranking; feedback bot; topic modelling/analysis; ML models exposed via APIs.

Leapfrog Technology

Kathmandu, Nepal

AI Intern

Oct 2018 – Mar 2019

- Predictive models for time-series weather and pollution forecasting (ARIMA and windowing-based).
- Model deployment and web-based interactive interface.

Department of Electronics & Computer Engineering (DOECE)

Lalitpur, Nepal

Student Research Assistant for UGC Research Grant Project

Jan 2018 – Sep 2018

- Assisted Prof. Dr. Nanda Bikram Adhikari on algorithm/tool research & implementation.
- Sensor data storage with GPS traces & data mining. Used RNN, RapidMiner, Hadoop, WebSocket, etc.

Education

Oslo Metropolitan University (OsloMet)

Oslo, Norway

Doctor of Philosophy (Ph.D.), at the Faculty of Technology, Art and Design (TKD)

May 2023 – May 2026

Thesis: Bridging Multimedia Modalities: Enhanced Multimodal AI Understanding and Intelligent Agents.

Coursework: Topics in AI and ML; Scientific Research Methods and Data Analysis in Engineering Science; Engineering Science and Ethics; Communicating Scientific Research (Simula Academy); Neural Methods in NLP (University of Oslo).

Research Interests: Multimodal AI; Vision-Language Models; Efficient Model Fine-Tuning; Uncertainty Quantification.

Thapathali Campus, Tribhuvan University

Kathmandu, Nepal

Master's Degree in Informatics & Intelligent Systems Engineering

Oct 2020 – Oct 2022

Full Scholarship for Academic Merit, Top of Class

Relevant coursework: Natural Language Processing; Computer Vision; Applied Machine Learning; Deep Learning; Computational Intelligence; Information Theory; Information Visualization; Optimization.

Pulchowk Campus, Tribhuvan University

Lalitpur, Nepal

Bachelor's Degree in Computer Engineering

Oct 2015 – Oct 2019

Full Scholarship for Academic Merit

Relevant coursework: Artificial Intelligence; Probability & Statistics; Data Mining; Big Data; Theory of Computation.

Technical Capabilities

Programming: Python, Android (Java/Kotlin), SQL, Bash, C/C++.

Libraries and tools: ML/DL, data analysis, numerical computing, visualization, web-frameworks, containerization, cloud, and backend-infrastructure ecosystems (including PyTorch, TensorFlow/Keras, NumPy/SciPy, Pandas, Scikit-learn, Matplotlib, OpenCV, Django, FastAPI, HuggingFace, LangChain, vLLM, PEFT/LoRA, Gradio, Streamlit, Git, Docker, Jupyter, etc.).

Languages: English (Fluent), Nepali (Native), Hindi (Advanced), Norwegian (Basic).

Academic Projects

MS Thesis: [AI-Based Soccer Game Summarization: From Video Highlights to Dynamic Text Summaries](#).

Best thesis in the department (by scores). In collaboration with the Department of Holistic Systems, **SimulaMet**.

MS Project: [Deep Learning-based Pose Estimation for Dystonia Score Prediction](#). In collaboration with **NAAMII**.

Bachelor's Thesis: [Assessment of Spatiotemporal Weather and Pollution Proxies with RNN and ARIMA](#).

Best Project Award by Department.

Bachelor's Project: [Human Facial Landmarks Detection using Multi-Layer Perceptron](#).

Selected Publications

2025: [Kvasir-VQA-x1: A Multimodal Dataset for Medical Reasoning and Robust MedVQA in Gastrointestinal Endoscopy](#). *MICCAI Workshop on Data Engineering in Medical Imaging*. LNCS, Springer.

2024: [Kvasir-VQA: A Text-Image Pair GI Tract Dataset](#). *VLM4Bio*, ACM Multimedia.

2024: [SoccerNet-Echoes: A Soccer Game Audio Commentary Dataset](#). *IEEE International Symposium on Multimedia*

2023: [Bridging Multimedia Modalities: Enhanced Multimodal AI Understanding and Intelligent Agents](#). *ICMI*, ACM.

2022: [Soccer Game Summarization using Audio Commentary, Metadata, and Captions](#). *Workshop on User-centric Narrative Summarization of Long Videos (NarSUM)*, ACM Multimedia. (Part of master's thesis work)

Awards & Certifications

NAAMII AI Winter School: [Full scholarship for participation](#). Topics: Deep Learning; Computer Vision; Statistics & Algorithms; Data Science; Optimization.

Hult Prize TU Winner (2017): [Winning team "AEGIS"](#). Presented a sustainable idea to empower refugees in Nepal using an e-commerce platform; presented at Hult Business School, Dubai.

Mentor: Global Venture Bootcamp, Asian Development Bank, Philippines — enhancing social ventures with emerging technologies.

Participant: IEEE AI & Big Data Workshop 2018; led multiple IEEE Student Chapter activities on campus.

Participant: UNICEF 1000 Golden Hours Code Camp (2017) — Best Prototype Award for mobile app idea.

References

Prof. Pål Halvorsen: Chief Research Scientist, SimulaMet; Professor, OsloMet; Adjunct Professor, UiO, Norway
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Dr. Michael Riegler: Head of AI, Simula Research Laboratory, Norway
michael@simula.no; +47 401 06 386.

Dr. Vajira Thambawita: Senior Research Scientist, Department of Holistic Systems (HOST), SimulaMet, Norway.
vajira@simula.no; +47 908 06 683.